



User Manual

Part 3: InfoCodex API

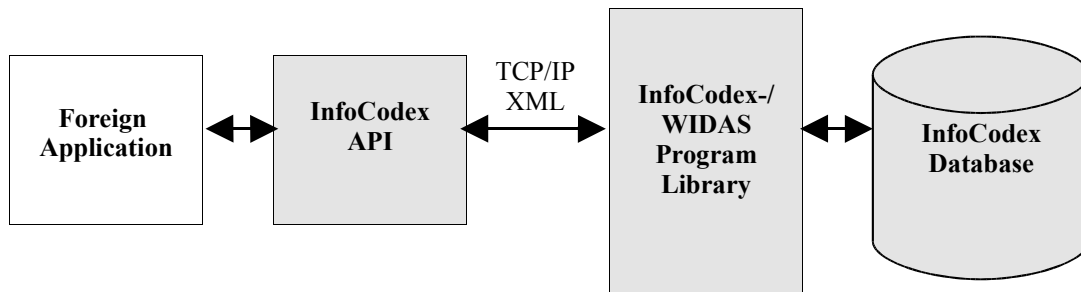
CONTENTS

User Manual.....	1
Part 3: InfoCodex API.....	1
1. Overview.....	5
2. Configuration.....	5
3. API Function ICApiFnc.....	6
4. Function Definitions.....	7
4.1 Function: <i>GetLicenceInfo</i>	7
4.2 Function: <i>SetClearingCode</i>	7
4.3 Function: <i>OpenSession</i>	7
4.4 Function: <i>CloseSession</i>	7
4.5 Function: <i>ExtractText</i>	8
4.6 Function: <i>AnalyzeDocument</i>	8
4.7 Function: <i>InsertDocument</i>	8
4.8 Function: <i>DeleteDocument</i>	9
4.9 Function: <i>GetDocument</i>	9
4.10 Function: <i>SearchDocument</i>	10
4.11 Function: <i>GetMap</i>	11
4.12 Function: <i>GetNeuron</i>	11
4.13 Function: <i>GetProjects</i> - Retrieve Collections.....	12
4.14 Function: <i>CreateProject</i> - Create Empty Collection.....	12
4.15 Function: <i>OpenProject</i> - Open Existing Collection.....	12
4.16 Function: <i>DeleteProject</i> - Delete Current Collection.....	13
4.17 Function: <i>EmptyProject</i> -Empty Current Collection (remove all documents).....	13
4.18 Function: <i>ReorganizeProject</i> - Reorganize Current Collection.....	13
4.19 Function: <i>GetProjectEnvironment</i>	15
4.20 Function: <i>SetProjectEnvironment</i>	15
4.21 Function: <i>GetUserGroups</i> - Return User Groups.....	15
4.22 Function: <i>GetSearchWords</i> - Return Words from Current Collection.....	15
4.23 Function: <i>AddDocuments</i>	16
5. XML Schema ic_capi.xsd (not yet adapted).....	16

6. Error Messages.....	23
<i>General Error Messages.....</i>	<i>23</i>
<i>Function: GetLicenceInfo.....</i>	<i>23</i>
<i>Function: SetClearingCode.....</i>	<i>23</i>
<i>Function: OpenSession.....</i>	<i>23</i>
<i>Function: SearchDocument.....</i>	<i>24</i>
<i>Function: ExtractText.....</i>	<i>24</i>
<i>Function: AnalyzeDocument.....</i>	<i>24</i>
<i>Function: InsertDocument.....</i>	<i>24</i>
<i>Function: DeleteDocument.....</i>	<i>24</i>
<i>Function: GetDocument.....</i>	<i>24</i>
<i>Function: CloseSession.....</i>	<i>25</i>
<i>Function: GetMap.....</i>	<i>25</i>
<i>Function: GetNeuron.....</i>	<i>25</i>
<i>Function: GetProjects.....</i>	<i>25</i>
<i>Function: OpenProject.....</i>	<i>25</i>
<i>Function: ReorganizeProject.....</i>	<i>25</i>
<i>Function: EmptyProject.....</i>	<i>25</i>
<i>Function: GetUserGroups.....</i>	<i>25</i>
<i>Function: AddDocuments.....</i>	<i>25</i>
<i>CreateProject.....</i>	<i>26</i>
<i>DeleteProject.....</i>	<i>26</i>
<i>GetProjectEnvironment.....</i>	<i>26</i>
<i>SetProjectEnvironment.....</i>	<i>26</i>
<i>GetSearchWords.....</i>	<i>26</i>
<i>GetDomains.....</i>	<i>26</i>

Version	Datum	Wer	Was
1.0	07.04.04	MHo	Basis version
1.1	10.05.04	MHo	-additional element 'UserLanguage' for 'OpenSessionIn' for user language introduced. -element 'Language' for document language of 'MetaData' moved into 'ContentData', because function 'ExtractText' cannot effectively determine the documents' language.
1.2	02.08.04	MHo	-additional element 'ThematicKey' for 'SearchDocumentIn' for thematic searches -element 'CreationDateUntil' corrected to 'CreationDateTo'
1.3	10.08.04	MHo	-additional function for map output
1.4	23.09.04	MHo	- additional function for map output extended/revised -new file type 'Notes'
1.5	23.11.04	MHo	- new file type 'Outlook' -new chapter 'Error messages'
1.6	25.11.04	MHo	-new error message 1022
1.7	23.12.04	MHo	-functions 'OpenSession', 'InsertDocument', 'SearchDocument','GetMap', 'GetNeuron' altered -new functions 'GetProjects', 'CreateProject', 'OpenProject', 'DeleteProject', 'EmptyProject', 'ReorganizeProject', 'GetProjectEnvironment', 'SetProjectEnvironment', 'GetUserGroups', 'GetSearchWords', 'AddDocuments'
1.8	12.01.05	MHo	new functions marked blue from V1.7 that have already been implemented in API V2.0.
1.9	14.02.05	MHo	new functionality in AddDocument
1.10	29.4.05	RHe	-new functions CreateProject, DeleteProject, SetProjectEnviroment, GetProjectEnviroment and GetSearchWords implemented. -additional elements Relevance and Neuronlist in function SearchDocument
1.11	11.5.05	MHo	file type 'Outlook' new 'MSG'

1. Overview



InfoCodex API Module <i>icapi.dll</i>	generic InfoCodex for embedding into foreign applications
InfoCodex / WIDAS-Program Library	diverse background processes, available to the InfoCodex / WIDAS functions.
InfoCodex DB	InfoCodex database

2. Configuration

The configuration data is stored in the Windows registry.
The administration of these entries is the responsibility of the foreign application.

Umgebungsvariablen:

Name	Meaning
IC	main InfoCodex directory with project subdirectories
WIDAS2	main directory for WIDAS programs

3. API Function ICApiFnc

```
int ICApiFnc ( char* InParam,
               int MaxBuf,
               char* OutParam,
               char* Msg);
```

Calling InfoCodex routines

Parameter	Description	IN/OUT
InParam	XML string with input parameters e.g.: <?xml version="1.0"?> <ICTelegram> <SearchDocumentIn> <SessionID>12345</SessionID> <FulltextKey>Vertrag 10023</FulltextKey> <SearchOption> <OrderBy>relevance</OrderBy> <MaxResults>20</MaxResults> </SearchOption> </SearchDocumentIn> </ICTelegram>	I
MaxBuf	maximum buffer size for output parameters	I
OutParam	XML string with output parameters e.g.: <?xml version="1.0"?> <ICTelegram> <SearchDocumentOut> <TotalFound>3</TotalFound> <DocumentList> <DocumentID>32f34f4r34r4f4</DocumentID> <DocumentID>fe4tr4gf54t5tg</DocumentID> <DocumentID>4g43t453f434f3</DocumentID> </DocumentList> </SearchDocumentOut> </ICTelegram>	O
Msg	string containing text error message (max. 1024 chars)	O

Return value

the return value of the called function.

0: error free execution of the function

>0: error code

It is the responsibility of the calling application to allocate sufficient memory for the output parameters. The size of the output buffer is supplied to the API in the parameter: MaxBuf.

For larger amounts of data, the XML parameter string can be used as a reference to XML files.

4. Function Definitions

The specification of the input/output parameters is based on the C API function ICApiFnc.

The XML syntax of the input/output parameters is defined by the file **ic_capi.xsd** (XML schema). There is an input/output XML structure for each function e.g. for the function SearchDocument: SearchDocumentIn/SearchDocumentOut.

4.1 Function: GetLicenceInfo

In	-		
Out	LicenceNumber	The installation's licence number	
	ComputerID	machine specific number	

4.2 Function: SetClearingCode

In	ClearingCode	licence clearing code	
Out	GoodThru	'valid until' date	

4.3 Function: OpenSession

In	ProjectID	unique collection number	optional
	UserName	user name (e.g. auch 'public')	
	Password	password (needed when authorization by InfoCodex is required)	optional
	UserLanguage	user language (e.g. for delivered keywords)	
Out	SessionID	unique ID number for an active session	

4.4 Function: CloseSession

In	SessionID	unique ID number of an active session	
Out	-		

4.5 Function: ExtractText

In	SessionID	unique ID number of an active session	
	DocumentName	filename and path of the document to be processed	
	ExtractName	filename and path of the text extract to be created	
Out	MetaData -Title -Author -FileType -CreationDate	metadata extracted from the document. (the possibility to obtain metadata from a document depends on the file type. Only those metadata fields are returned which are available from that file type.)	
	+ a separate file 'ExtractName'	text	

4.6 Function: AnalyzeDocument

In	SessionID	unique ID number of an active session	
	ExtractName	filename and path of the text document to analyze	
	MetaData -Title -Author -FileType -CreationDate	metadata of the document (metadata can be useful in the creation of keywords e.g. document title.)	optional
	MaxKeywords	number of keywords to return 0-20 (default 20)	optional
Out	ContentData -NeuronCoord -NoOfWords -Language -KeywordList -Keyword	results of the content analysis: - coordinates of the best fitting neuron (e.g. M7) - total words in the document - detected document language - keyword list with up to MaxKeywords of automatically determined entries	

4.7 Function: InsertDocument

In	SessionID	unique ID number of an active session	
	DocumentID or DocumentName	unique identification of the document (e.g. GUID or filename and path of document) or DocumentName	
	ExtractName	filename and path of the text document to be analyzed	
	MetaData -Title -Author -FileType -CreationDate	Document's metadata	optional
	UserGroups -UserGroupID	access restrictions	optional
	PreparedCollection	documents are not directly loaded into InfoCodex, but stored as a prepared .TXT file collection. The actual import occurs with the function 'AddDocuments'.	optional
	ContentOutput -MaxKeywords	output of 'ContentData' desired -maybe with input keywords	optional

Out	DocumentID	unique identification of the document in the case where input DocumentName is used	optional
	ContentData -NeuronCoord -NoOfWords -Language -KeywordList -Keywords	results of the content analysis: - coordinates of the best fitting neuron - total words in the document -detected document language - keyword list with up to MaxKeywords of automatically determined entries	optional

4.8 Function: DeleteDocument

In	SessionID	unique ID number of an active session	
	DocumentID	unique identification of the document (e.g. GUID or filename and path of document)	
Out	-		

4.9 Function: GetDocument

In	SessionID	unique ID number of an active session	
	DocumentID	unique identification of the document (e.g. GUID or filename and path of document)	
	MaxKeywords	number of keywords to return 0-20 (default 20)	optional
Out	MetaData -Title -Author -FileType -CreationDate	document metadata	
	ContentData -NeuronCoord -NoOfWords -Language -KeywordList -Keyword	results of the content analysis: - coordinates of the best fitting neuron - total words in the document -detected document language - keyword list with up to MaxKeywords of automatically determined entries	

4.10 Function: SearchDocument

In	SessionID	unique ID number of an active session	
	SearchKey -ThematicKey -Text -TextFile -FulltextKey -MetaDataKey -Title -AuthorList -LanguageList -FileTypeList - CreationDateFrom -CreationDateUntil	search terms with 1 or more search methods: -thematic search by input of a text block or text file. - full text search (syntax as in the InfoCodex input field) - index search with 1 or more metadata, as - title (start) - list of authors - list of language codes - list of file types - creation date from / to	optional optional optional
	TopicOutput	output of 'TopicList' desired	optional
	NeuronOutput	output of 'NeuronList' as HeatMap desired	optional
	DocumentOutput -MaxResults -OrderBy -MetaOutput -ContentOutput -MaxKeywords	output of 'DocumentList' desired - maximum number of hits (default 5000) - sort order (relevance, size, date) - output of 'MetaData' desired - output of 'ContentData' desired - input of the keywords to be retrieved 0-20 (default 20)	optional
	OutputFile	output not in OutParam but as separate files	optional
Out	TotalDocuments	total number of hits	
	TopicList -TopicData -TopicID -TopicText	list of thematic areas with data per thematic area: - unique ID - label	optional optional
	NeuronList -NeuronData -NeuronCoord -TopicID -NoOfDocuments -KeywordList -Keyword -Relevance	list of the neurons with data per neuron: - coordinates - assignment to thematic area - total documents - list with up to 5 neuron keywords - relevance in %	optional optional
	DocumentList -DocumentData -DocumentID -Relevance -MetaData -Title -Author -FileType -CreationDate -ContentData -NeuronCoord -NoOfWords -Language -KeywordList -Keyword	list of found documents with data per document: - unique ID of the document - relevance in % when OrderBy=relevance - document's metadata - results of the content analysis - neuron coordinates - total words in document - detected document language - keyword list with up to 20 automatically determined keywords	optional optional optional optional optional
as	a separate file 'OutputFile'	file containing output, in the case where option 'OutputFile' was requested.	optional

4.11 Function: GetMap

In	SessionID	unique ID number of an active session	
	OutputFile	output not in OutParam but as separate file	optional
Out	TotalDocuments	total documents	
	Structurability	structurability in %	
	TopicList -TopicData -TopicID -TopicText	list of thematic areas with data per thematic area: - uniqueID - label	optional
	NeuronList -NeuronData -NeuronCoord -TopicID - NoOfDocuments -KeywordList -Keyword	list of neurons with data per neuron: - coordinates - assignment to thematic area - total documents - list with up to 5 neuron keywords	optional
as	A separate file 'OutputFile'	file with output, in the case where 'OutputFile' was specified.	optional

4.12 Function: GetNeuron

In	SessionID	unique ID number of an active session	
	NeuronCoord	neuron coordinates	
	HeatMap	neuron HeatMap specification from a previous search	optional
	TopicOutput	'TopicData' output requested	optional
	NeuronOutput	'NeuronData' output requested	optional
	DocumentOutput -MaxResults -OrderBy -MetaOutput -ContentOutput -MaxKeywords	output of 'DocumentList' requested - maximum number of results to return - sort order (relevance, size, date) - output of 'MetaData' requested - output of 'ContentData' requested - maximum no. of keywords to return 0-20 (default 20)	optional
	OutputFile	output in separate file	optional
Out	TopicData -TopicID -TopicText	data of topic area: - unique ID - label	optional
	NeuronData -NeuronCoord -TopicID -NoOfDocuments -KeywordList -Keyword	data per neuron: - coordinates - ID of topic area - total documents - list of up to 5 neuron keywords	optional

	DocumentList -DocumentData -DocumentID -Relevance -MetaData -Title -Author -FileType -CreationDate -ContentData -NeuronCoord -NoOfWords -Language -KeywordList -Keyword	list of retrieved documents with data per document: - unique document ID - relevance in % by OrderBy=relevance - document metadata - results of content analysis - neuron coordinates - total words in document - detected document language - list of up to 20 automatically determined keywords keywords	optional optional optional optional optional
as	A separate file 'OutputFile'	file containing output if option 'OutputFile' was requested	optional

4.13 Function: GetProjects - Retrieve Collections

In	SessionID	unique ID number of an active session	
Out	ProjectList -ProjectData -ProjectID -ProjectName -ProjectState	List of available collections with data per collection: - unique ID - label - status ('processing' or 'reconstruct')	optional optional

4.14 Function: CreateProject - Create Empty Collection

In	SessionID	unique ID number of an active session	
	ProjectEnvironment -ProjectID -ProjectNames -German -English -French -Italian -UserGroups -UserGroupID	unique collection number collection name in the different languages access restrictions	optional at least in one language
Out	ProjectID	unique collection number generated by InfoCodex in the case where no ProjectID was given as input	

4.15 Function: OpenProject - Open Existing Collection

In	SessionID	unique ID number of an active session	
	ProjectID	unique collection number	
	TopicOutput	'TopicList' output requested	optional
	NeuronOutput	'NeuronList' output requested	optional
	OutputFile	output to separate file	optional
Out	TotalDocuments	total number of documents	

	Structurability	structurability in %	
	TopicList -TopicData -TopicID -TopicText	list of topic areas with data per topic area: - unique ID - label	optional optional
	NeuronList -NeuronData -NeuronCoord -TopicID -NoOfDocuments -KeywordList -Keyword	list of neurons with data per neuron: - coordinates - topic area ID - total documents - list of up to 5 neuron keywords	optional optional
as	a separate file 'OutputFile'	file containing output if option 'OutputFile' was requested.	optional

4.16 Function: DeleteProject - Delete Current Collection

In	SessionID	unique ID number of an active session	
	ProjectID	number of current collection (as control)	
Out	-		

4.17 Function: EmptyProject -Empty Current Collection (remove all documents)

In	SessionID	unique ID number of an active session	
	ProjectID	number of current collection (as control)	
	ExternalKey	parameter if an own DocumentID by InsertDocument will be given	optional
Out	-		

4.18 Function: ReorganizeProject - Reorganize Current Collection

In	SessionID	unique ID number of an active session	
	ProjectID	number of current collection (as control)	
	TopicOutput	'TopicList' output requested	optional
	NeuronOutput	'NeuronList' output requested	optional
	OutputFile	output in a separate file	optional
Out	TotalDocuments	total documents	
	Structurability	structurability in %	
	TopicList -TopicData -TopicID -TopicText	list of topic areas with data per topic area: - unique ID - label	optional optional
	NeuronList -NeuronData -NeuronCoord -TopicID -NoOfDocuments -KeywordList -Keyword	list per neuron with data per neuron: - coordinates - topic area ID - total documents - list with up to 5 neuron keywords	optional optional

<i>as</i>	a separate file 'OutputFile'	file containing output if option 'OutputFile' was requested.	optional
-----------	---------------------------------	---	----------

4.19 Function: *GetProjectEnvironment*

In	SessionID	unique ID number of an active session	
Out	ProjectEnvironment -ProjectID -ProjectNames -German -English -French -Italian -UserGroups -UserGroupID	unique collection number collection name in the different languages access restrictions	

4.20 Function: *SetProjectEnvironment*

In	SessionID	unique ID number of an active session	
	ProjectEnvironment -ProjectID -ProjectNames -German -English -French -Italian -UserGroups -UserGroupID	unique collection number (as control) collection name in the different languages access restrictions	optional optional
Out	-		

4.21 Function: *GetUserGroups - Return User Groups*

In	SessionID	unique ID number of an active session	
Out	UserGroupList -UserGroupData -UserGroupID -UserGroupName	list of available user groups with data per user group: - unique ID - name of group	optional

4.22 Function: *GetSearchWords - Return Words from Current Collection*

In	SessionID	unique ID number of an active session	
	SearchWord	search term for word (beginning)	
	MaxWords	maximum no. words to return	optional
	OutputFile	output in separate file	optional
Out	WordList -Word	list of available words in the collection	optional
as	a separate file 'OutputFile'	file containing output if option 'OutputFile' was requested.	optional

4.23 Function: AddDocuments

In	SessionID	unique ID number of an active session	
	PreparedCollection -DeleteCollection	load prepared TXT file collection - delete TXT files afterwards	optional
	NotesConnect -IndexFile -DriveMapping -From -To	import from NotesConnect indexfile - file and pathname of IDX file - map drives of document pathnames in IDX file	optional
	UserGroups -UserGroupID	access restrictions	optional
	ActivateOCR	activate OCR processing	optional
	TopicOutput	'TopicList' output requested	optional
	NeuronOutput	'NeuronList' output requested	optional
	OutputFile	output in separate file	optional
Out	TotalDocuments	total documents	
	Structurability	structurability in %	
	TopicList -TopicData -TopicID -TopicText	list of topic areas with data per topic area: -unique ID - label	optional optional
	NeuronList -NeuronData -NeuronCoord -TopicID -NoOfDocuments -KeywordList -Keyword	list of neurons with data per neuron: - coordinates - topic area ID - total documents - list with up to 5 neuron keywords	optional optional
as	a separate file 'OutputFile'	file containing output if option 'OutputFile' was requested.	optional

5. XML Schema ic_capi.xsd (not yet adapted)

```

<!-- ===== -->
<!-- ic_capi.xsd    XML Schema for InfoCodex C-API -->
<!-- -->
<!-- MHO -->
<!-- 7.5.04/ 2.8.04/ 10.8.04/ 29.9.04 -->
<!-- -->
<!-- ===== -->
<schema targetNamespace="http://www.infocodex.com/capi"
xmlns:icapi="http://www.infocodex.com/capi" xmlns="http://www.w3.org/2001/XMLSchema"
elementFormDefault="qualified">
  <annotation>
    <documentation>XML Schema for InfoCodex C-API</documentation>
  </annotation>
  <complexType name="ICTelegram">
    <choice>
      <element name="GetLicenceInfoIn" type="icapi:GetLicenceInfoIn"/>
      <element name="GetLicenceInfoOut" type="icapi:GetLicenceInfoOut"/>
      <element name="SetClearingCodeIn" type="icapi:SetClearingCodeIn"/>
      <element name="SetClearingCodeOut" type="icapi:SetClearingCodeOut"/>
      <element name="OpenSessionIn" type="icapi:OpenSessionIn"/>
      <element name="OpenSessionOut" type="icapi:OpenSessionOut"/>
      <element name="CloseSessionIn" type="icapi:CloseSessionIn"/>
      <element name="CloseSessionOut" type="icapi:CloseSessionOut"/>
      <element name="ExtractTextIn" type="icapi:ExtractTextIn"/>
      <element name="ExtractTextOut" type="icapi:ExtractTextOut"/>
      <element name="AnalyzeDocumentIn" type="icapi:AnalyzeDocumentIn"/>
    </choice>
  </complexType>
</schema>

```



```

        <element name="AnalyzeDocumentOut" type="icapi:AnalyzeDocumentOut"/>
        <element name="InsertDocumentIn" type="icapi:InsertDocumentIn"/>
        <element name="InsertDocumentOut" type="icapi:InsertDocumentOut"/>
        <element name="DeleteDocumentIn" type="icapi:DeleteDocumentIn"/>
        <element name="DeleteDocumentOut" type="icapi:DeleteDocumentOut"/>
        <element name="GetDocumentIn" type="icapi:GetDocumentIn"/>
        <element name="GetDocumentOut" type="icapi:GetDocumentOut"/>
        <element name="SearchDocumentIn" type="icapi:SearchDocumentIn"/>
        <element name="SearchDocumentOut" type="icapi:SearchDocumentOut"/>
        <element name="GetMapIn" type="icapi:GetMapIn"/>
        <element name="GetMapOut" type="icapi:GetMapOut"/>
        <element name="GetNeuronIn" type="icapi:GetNeuronIn"/>
        <element name="GetNeuronOut" type="icapi:GetNeuronOut"/>
    </choice>
</complexType>
<!-- Definition of Function Parameters -->
<complexType name="GetLicenceInfoIn"/>
<complexType name="GetLicenceInfoOut">
    <sequence>
        <element name="LicenceNumber" type="unsignedInt"/>
        <element name="ComputerID" type="string"/>
    </sequence>
</complexType>
<complexType name="SetClearingCodeIn">
    <sequence>
        <element name="ClearingCode" type="string"/>
    </sequence>
</complexType>
<complexType name="SetClearingCodeOut">
    <sequence>
        <element name="GoodThru" type="date"/>
    </sequence>
</complexType>
<complexType name="OpenSessionIn">
    <sequence>
        <element name="ProjectID" type="unsignedInt"/>
        <element name="UserName" type="string"/>
        <element name="Password" type="string" minOccurs="0"/>
        <element name="UserLanguage" type="icapi:Language"/>
    </sequence>
</complexType>
<complexType name="OpenSessionOut">
    <sequence>
        <element name="SessionID" type="icapi:SessionID"/>
    </sequence>
</complexType>

```

```

<complexType name="CloseSessionIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
  </sequence>
</complexType>
<complexType name="CloseSessionOut"/>
<complexType name="ExtractTextIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="DocumentName" type="string"/>
    <element name="ExtractName" type="string"/>
  </sequence>
</complexType>
<complexType name="ExtractTextOut">
  <sequence>
    <element name="MetaData" type="icapi:MetaData"/>
  </sequence>
</complexType>
<complexType name="AnalyzeDocumentIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="ExtractName" type="string"/>
    <element name="MetaData" type="icapi:MetaData" minOccurs="0"/>
    <element name="MaxKeywords" type="icapi:MaxKeywords" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="AnalyzeDocumentOut">
  <sequence>
    <element name="ContentData" type="icapi:ContentData"/>
  </sequence>
</complexType>
<complexType name="InsertDocumentIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="DocumentID" type="string"/>
    <element name="ExtractName" type="string"/>
    <element name="MetaData" type="icapi:MetaData" minOccurs="0"/>
    <element name="InsertOption" type="icapi:InsertOption" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="InsertDocumentOut">
  <sequence>
    <element name="ContentData" type="icapi:ContentData" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="DeleteDocumentIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="DocumentID" type="string"/>
  </sequence>
</complexType>
<complexType name="DeleteDocumentOut"/>
<complexType name="GetDocumentIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="DocumentID" type="string"/>
    <element name="MaxKeywords" type="icapi:MaxKeywords" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="GetDocumentOut">
  <sequence>
    <element name="MetaData" type="icapi:MetaData"/>
    <element name="ContentData" type="icapi:ContentData"/>
  </sequence>
</complexType>
<complexType name="SearchDocumentIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="ThematicKey" type="icapi:ThematicKey" minOccurs="0"/>
    <element name="FulltextKey" type="string" minOccurs="0"/>
    <element name="MetaDataKey" type="icapi:MetaDataKey" minOccurs="0"/>
    <element name="SearchOption" type="icapi:SearchOption" minOccurs="0"/>
    <element name="OutputFile" type="string" minOccurs="0"/>
  </sequence>
</complexType>

```

```

<complexType name="SearchDocumentOut">
  <sequence>
    <element name="TotalFound" type="unsignedLong" minOccurs="0"/>
    <element name="DocumentList" type="icapi:DocumentList" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="GetMapIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="OutputFile" type="string" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="GetMapOut">
  <sequence>
    <element name="TopicList" type="icapi:TopicList" minOccurs="0"/>
    <element name="NeuronList" type="icapi:NeuronList" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="GetNeuronIn">
  <sequence>
    <element name="SessionID" type="icapi:SessionID"/>
    <element name="NeuronCoord" type="string"/>
    <element name="NeuronOption" type="icapi:NeuronOption" minOccurs="0"/>
    <element name="OutputFile" type="string" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="GetNeuronOut">
  <sequence>
    <element name="TopicText" type="string" minOccurs="0"/>
    <element name="NeuronData" type="icapi:NeuronData" minOccurs="0"/>
    <element name="DocumentList" type="icapi:DocumentList" minOccurs="0"/>
  </sequence>
</complexType>
<!-- Definition of Subtypes -->
<simpleType name="SessionID">
  <restriction base="unsignedLong"/>
</simpleType>
<simpleType name="MaxKeywords">
  <restriction base="unsignedInt">
    <minInclusive value="0"/>
    <maxInclusive value="20"/>
  </restriction>
</simpleType>
<simpleType name="Language">
  <restriction base="language">
    <enumeration value="en"/>
    <enumeration value="de"/>
    <enumeration value="fr"/>
    <enumeration value="it"/>
  </restriction>
</simpleType>
<simpleType name="OrderBy">
  <restriction base="string">
    <enumeration value="relevance"/>
    <enumeration value="size"/>
    <enumeration value="date"/>
  </restriction>
</simpleType>
<simpleType name="FileType">
  <restriction base="string">
    <enumeration value="ASCII"/>
    <enumeration value="Word"/>
    <enumeration value="HTML"/>
    <enumeration value="PDF"/>
    <enumeration value="E-Mail"/>
    <enumeration value="PowerPoint"/>
    <enumeration value="Excel"/>
    <enumeration value="Binary Image"/>
    <enumeration value="PS"/>
    <enumeration value="XML"/>
    <enumeration value="RTF"/>
    <enumeration value="Notes"/>
    <enumeration value="MSG"/>
  </restriction>
</simpleType>

```



```

<complexType name="KeywordList">
  <sequence>
    <element name="Keyword" type="string" minOccurs="0" maxOccurs="20"/>
  </sequence>
</complexType>
<complexType name="AuthorList">
  <sequence>
    <element name="Author" type="string" minOccurs="0"
maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="LanguageList">
  <sequence>
    <element name="Language" type="icapi:Language" maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="FileTypeList">
  <sequence>
    <element name="FileType" type="icapi:FileType" maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="DocumentList">
  <sequence>
    <element name="DocumentID" type="string" minOccurs="0"
maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="NeuronList">
  <sequence>
    <element name="NeuronData" type="icapi:NeuronData" minOccurs="0"
maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="TopicList">
  <sequence>
    <element name="TopicData" type="icapi:TopicData" minOccurs="0"
maxOccurs="unbounded"/>
  </sequence>
</complexType>
<complexType name="ContentOutput">
  <all>
    <element name="MaxKeywords" type="icapi:MaxKeywords" minOccurs="0"/>
  </all>
</complexType>
<complexType name="InsertOption">
  <all>
    <element name="ContentOutput" type="icapi:ContentOutput"
minOccurs="0"/>
  </all>
</complexType>
<complexType name="SearchOption">
  <all>
    <element name="MaxResults" type="unsignedLong" minOccurs="0"/>
    <element name="OrderBy" type="icapi:OrderBy" minOccurs="0"/>
  </all>
</complexType>
<complexType name="NeuronOption">
  <all>
    <element name="NoTopicText" type="unsignedByte" minOccurs="0"/>
    <element name="NoNeuronData" type="unsignedByte" minOccurs="0"/>
    <element name="NoDocumentList" type="unsignedByte" minOccurs="0"/>
    <element name="MaxResults" type="unsignedLong" minOccurs="0"/>
    <element name="OrderBy" type="icapi:OrderBy" minOccurs="0"/>
  </all>
</complexType>
<complexType name="ThematicKey">
  <choice>
    <element name="Text" type="string" minOccurs="0"/>
    <element name="TextFile" type="string" minOccurs="0"/>
  </choice>
</complexType>

```

```

<complexType name="MetaDataKey">
  <sequence>
    <element name="Title" type="string" minOccurs="0"/>
    <element name="AuthorList" type="icapi:AuthorList" minOccurs="0"/>
    <element name="LanguageList" type="icapi:LanguageList" minOccurs="0"/>
    <element name="FileTypeList" type="icapi:FileTypeList" minOccurs="0"/>
    <element name="CreationDateFrom" type="date" minOccurs="0"/>
    <element name="CreationDateTo" type="date" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="MetaData">
  <sequence>
    <element name="Title" type="string" minOccurs="0"/>
    <element name="Author" type="string" minOccurs="0"/>
    <element name="FileType" type="icapi:FileType" minOccurs="0"/>
    <element name="CreationDate" type="date" minOccurs="0"/>
  </sequence>
</complexType>
<complexType name="ContentData">
  <sequence>
    <element name="NeuronCoord" type="string"/>
    <element name="NoOfWords" type="unsignedLong"/>
    <element name="Language" type="icapi:Language"/>
    <element name="KeywordList" type="icapi:KeywordList"/>
  </sequence>
</complexType>
<complexType name="NeuronData">
  <sequence>
    <element name="NeuronCoord" type="string"/>
    <element name="TopicID" type="unsignedLong"/>
    <element name="NoOfDocuments" type="unsignedLong"/>
    <element name="KeywordList" type="icapi:KeywordList"/>
  </sequence>
</complexType>
<complexType name="TopicData">
  <sequence>
    <element name="TopicID" type="unsignedLong"/>
    <element name="TopicText" type="string"/>
  </sequence>
</complexType>
<!-- Needed by XML Instance Generator -->
<element name="ICTelegram" type="icapi:ICTelegram"/>
</schema>

```

6. Error Messages

Code	Text	Additional Information in ()
	General Error Messages	
	System relevant:	
1	Error open server socket	Code 'server_socket'
2	Invalid XML tag <SessionID>	
3	Invalid XML telegram	XML Input
4	Invalid SessionID	SessionID
5	Error sending CMD to server	Code 'tcp_send'
6	Error receiving CMD from server	Code 'tcp_recv'
7	Invalid CMD acknowledge received from server	String 'tcp_recv'
8	Invalid CMD acknowledge received from server	String 'tcp_recv'
9	Output string exceeds buffer size	Required buffer size
10	Error starting detached process	Code-Cmd 'RunCGI'
11	No accept on socket	
12	Error receiving session PID	Code 'tcp_recv'
13	Invalid session acknowledge	String 'tcp_recv'
14	Invalid <OutputFile> tag	XML string
	Session relevant:	
1020	Cannot open file	Name of file
1021	Cannot open file	Name of file
1022	Initial load of this project was not regularly terminated	
1025	Could not open file to save nodes	Filename
1026	Could not open file to restore nodes	Filename
1030	Background process was terminated	Status message
1031	Background process was terminated	
1100	Invalid command received	Cmd string
1101	Invalid XML telegram	XML Input
1102	Initial load not regularly completed -> incremental load not possible	
1103	No open project	
1104	Invalid metadata structure	Metadata
1999	Unexpected error closed session	Debug information
	Function: GetLicenceInfo	
20	Cannot open project table	
	Function: SetClearingCode	
25	Invalid XML tag <SetClearingCodeIn>	"<ClearingCode>"
26	Cannot open project table	Name of project table
27	Invalid registration	String 'RegisterMe'
28	Cannot open project table	Name of project table
	Function: OpenSession	
30	Invalid XML tag <OpenSessionIn>	Invalid tags
31	No more sessions available	
1001	Limit of simultaneous users has been reached	Max concurrent users
1002	Infocodex licence not valid on this machine	
1003	Invalid password	
1004	Invalid username	Username
1005	Licence not longer valid	
1006	Invalid project	ProjectID
1007	Project directory could not be created	
1008	Invalid ProjectID	ProjectID

1009	Project is not yet initialized	ProjectID
1015	Login not successfull	
1016	No access to filesystem security	FSSecurity Errormessage
1017	Invalid username in IC domain	Username
1027	Access to project not allowed	ProjectID

	Function: SearchDocument	
1130	Invalid XML tag <SearchDocumentIn>	Invalid tags
1131	Output string limited to 32k -> use option <OutputFile>	
1132	Error writing XML file	Name and line of output file
1135	Error preparing search file	Name and line of search file
	Function: ExtractText	
1010	Error starting detached process	Code-Cmd 'RunCGI'
1011	No accept on socket	
1012	Error receiving spider agent PID	Code 'tcp_recv'
1013	Error message: spider agent server	
1014	Invalid spider agent acknowledge	String 'tcp_recv'
1138	Invalid directory for extract file	ExtractName
1139	Invalid extract filename	ExtractName
1140	Invalid XML tag <ExtractTextIn>	Invalid tags
1141	Document not found	DocumentName
1143	Error open infocodex socket	Code 'server socket'
1144	Error sending CMD to icadd2	Code 'tcp_send'
1145	Error receiving CMD from icadd2	Code 'tcp_recv'
1146	Invalid CMD acknowledge received from icadd2	String 'tcp_recv'
1147	Invalid CMD acknowledge received from icadd2	String 'tcp_recv'
1148	Invalid FileType received from icadd2	FileType
1149	Invalid DATA received from icadd2	Metadata
2001	Environment variables are not defined	
2002	No temp dir specified	
2003	No prj dir specified	
2100	Invalid command received	String 'tcp_recv'
2101	Invalid command arguments received	String 'tcp_recv'
2102	Convert2Txt not successfull	Format specific error code
2999	Unexpected error closed icadd2	Debug information
	Function: AnalyzeDocument	
1150	Invalid XML tag <AnalyzeDocumentIn>	Invalid tags
1151	Extract file not found	ExtractName
	Function: InsertDocument	
1160	Invalid XML tag <InsertDocumentIn>	Invalid tags
1161	Extract file not found	ExtractName
1162	Not authorized to insert document	
1163	Not enough neurons (documents)	Rows in map.tb
1168	DocumentID not unique	DocumentID
1169	Function not available on this project	"InsertDocument"
	Function: DeleteDocument	
1170	Invalid XML tag <DeleteDocumentIn>	Invalid tags
1171	Document not found	DocumentID
1172	Not authorized to delete document	DocumentID
	Function: GetDocument	
1180	Invalid XML tag <GetDocumentIn>	Invalid tags
1181	Document not found	DocumentID

1182	Document not found	DocumentID
1183	Document not found	DocumentID
1184	Not authorized to view document	DocumentID
	Function: CloseSession	
1190	Invalid XML tag <CloseSessionIn>	"<SessionID>"
1191	Incorrect SessionID	SessionID/PID
	Function: GetMap	
1200	Output string limited to 32k -> use option <OutputFile>	
1202	Error writing XML file	Name and line of output file
	Function: GetNeuron	
1211	Invalid XML tag <GetNeuronIn>	Invalid tags
1212	Neuron not found	NeuronCoord
1213	Output string limited to 32k -> use option <OutputFile>	
1214	Error writing XML file	Name and line of output file
	Function: GetProjects	
	Function: OpenProject	
1220	Invalid XML tag <OpenProjectIn>	Invalid tags
1221	Invalid project	ProjectID
1222	Access to project not allowed	ProjectID
1223	Project opening not successfull	
1224	Error writing XML file	Name and line of output file
	Function: ReorganizeProject	
1230	Invalid XML tag <ReorganizeProjectIn>	Invalid tags
1231	ProjectID belongs not to current project	ProjectID
1232	Not authorized to reorganize project	
1233	Not possible for current project status	Project status
1234	Input file(s) are missing	Filename(s)
1235	Not possible for this project	
1236	Error starting detached process	Code-Cmd 'RunCGI'
1237	Reopen of project not successfull	
1238	Error writing XML file	Name and line of output file
	Function: EmptyProject	
1240	Invalid XML tag <EmptyProject>	Invalid tags
1241	ProjectID belongs not to current project	ProjectID
1242	Not possible for current project status	Project status
1243	Cleaning tables not successfull	Errorcode ICAPrj
1244	Reopening of project not successful	
	Function: GetUserGroups	
1251	Can not open user group table	Tablename
	Function: AddDocuments	
1260	Invalid XML tag <AddDocumentsIn>	Invalid tags
1261	Index file not found	Name index file
1262	Not authorized to administrate project	
1263	Not possible for current project status	Project status / Error code
1265	Error starting detached process	Code-Cmd 'RunCGI'
1266	Reopen of project not successfull	
1267	Error writing jobs file	Name jobs file
1268	Error writing XML file	Name and line of output file

	CreateProject	
1271	ProjectID already in use	ProjectID
1272	ProjectID has wrong format	ProjectID
1273	Missing tag	Tag name
1275	Cannot open table	Table name
1276	User group does not exist	User group ID
1277	Not authorized to create Project	
	DeleteProject	
1280	Missing tag	Tag name
1281	ProjectID has wrong format	ProjectID
1282	ProjectID is not identical with current project	
1283	Not authorized to delete project	
1284	Project does not exist	ProjectID
	GetProjectEnvironment	
1290	Project does not exist	ProjectID
1291	Not authorized for this function	
	SetProjectEnvironment	
1300	Not authorized for this function	
1301	Cannot open table	Table name
1302	ProjectID has wrong format	ProjectID
1303	Missing tag	Tag name
1304	User group does not exist	User group ID
1305	ProjectID is not identical with current project	
1306	Project does not exist	ProjectID
	GetSearchWords	
1310	Missing tag	Tag name
1311	MaxWords has wrong format	MaxWords
	GetDomains	
1320	Cannot open domain table	Table name